

Determining the Crystalline Structure of Metals through X-Ray Diffraction

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Abstract

This experiment set out to measure X-ray diffraction patterns of various solid samples in order to describe their crystalline structure level. A pre-calibrated Siemen D5000 X-Ray diffractometer was used to characterize the structure of Copper, Nickel (Ni), CuNi alloy, Titanium (Sn), Lead (Pb), PbSn alloy and two mystery samples A and B. Bragg's law was used to obtain the distances of various diffraction planes, which was related to the Miller indices of the planes. It was found that Copper, Nickel and the alloys are Cubic system with face centered (FCC) Bravais lattices ((and a lattice parameter of [...])); Lead was also found to be FCC, while the Tin sample was a body-centered tetragonal, and the alloys are Tetragonal systems with [P/I] lattices ((and parameters of)); sample A is a cubic system with a face-centered lattice whereas B is also a cubic system, but with a primitive lattice (PC).

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1 Introduction

A solid can serve as a diffraction grating for X-rays of wavelengths similar to the lengths of inter-atomic distances. Similarly to how light diffracts, X-rays can also diffract off of atoms. Atoms within solids will arrange themselves in symmetries, formally called cells, given by one of one of fourteen Bravais lattices, with each lattice grouped into one of seven crystal systems. These Bravais lattices govern along which diffraction planes atoms may align and Miller indices are conventionally used to index the various planes. Appendix 6.1 shows the different Bravais lattices. If an X-ray is sent to a solid with an angle θ such that constructive interference occurs at a detector placed 2θ with respect to the incident ray, a peak in intensity of photons may be observed. From this, one can obtain a relation between the inter-planar distance d , the incident angle θ , and the wavelength λ .

$$n\lambda = 2d \sin(\theta) \tag{1}$$

with $n = 1$ for our purposes. Equation 1 is formally called Bragg's Law.

Using these observed peaks and (1), the distance d between the parallel diffraction planes can be obtained and related to the Bravais lattice and Miller indices by various equations. For a cubic system, this relation takes the form of

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2} \tag{2}$$

where h, k, l are the Miller Indices and a is the length of the cube edge. Similar equations for different crystal systems are given in 6.1. Additionally, selection rules exist for each Bravais lattice, which allow only certain diffraction planes to be observed. For instance, while any plane is allowed for a primitive cubic lattice, only planes with all even or all odd Miller indices are allowed for a face-centered cubic lattice. These selection rules are outlined for cubic and tetragonal systems in Appendix 6.1.

2 Methods

2.1 Instrument and Data Acquisition

The apparatus consisted of a pre-calibrated Siemens D5000 diffractometer. The X-ray tube used Copper K_α radiation. Two sets of filters of 0.6 mm were present on the X-ray tube and the detector. During data acquisition, the diffractometer was set to 40 kV and 40 mA. A computer connected to the apparatus was used to specify the measurement parameters and obtain the data from the instrument. A picture of the instrument is shown in Figure 1

Data acquisition consisted in mounting the sample on the mounting plate, opening the parameter file in the computer software and executing the job. Once the job is executed, the detector rotates about the mounting plate to be at an angle of $2\theta_0 = 5.00^\circ$ with respect to the incident rays and the mounting plate rotates about itself to make an angle of θ_0 . The detector and mounting plate rotate with angle increments of $2\Delta\theta = 0.05$ and $\Delta\theta$, respectively, with 2.00 s at each step until reaching $2\theta = 110.00^\circ$.

Measurements were taken for samples of Copper, Nickel, Tin, Lead; 25%, 50% and 75% alloys of Copper-Nickel and Tin-lead, and two mystery samples A and B. Three trials were initially done for Copper and Nickel, and two trials were done for the other samples. This is because it was found that the uncertainties associated to statistical fluctuations of two measurements were smaller than that associated with noise in a single data plot, expressed as standard deviation of a fitted gaussian. Therefore, taking a second trial reduced the uncertainty in the mean value of d for a given peak, but taking a third trial did not reduce it significantly.

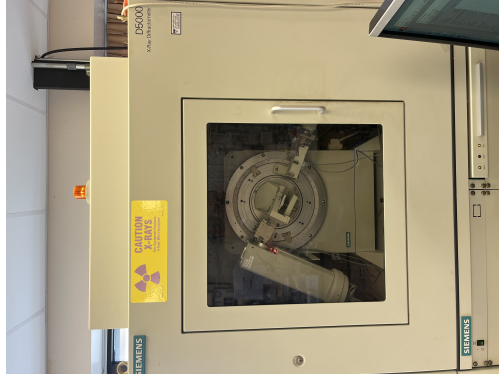


Figure 1: Picture of the Siemens D5000 diffractometer used for the experiment.

2.2 Data Analysis

The measured intensity was read as a function of 2θ for each sample. Candidate diffraction peaks were identified from a smoothed intensity trace using a local-maximum prominence threshold. These parameters were varied from sample to sample based on the characteristics of the diffraction patterns. Smoothing was used only for candidate detection; the peak fit used the original measured intensities.

All candidate peaks in a sample plot were fitted simultaneously with a sum of Gaussian functions and a linear background (meant to account for a baseline count), $I_{\text{fit}}(2\theta) = m(2\theta) + b + \sum_i A_i \exp[-(2\theta - \mu_i)^2 / (2\sigma_i^2)]$. The initial peak centers were the detected positions, and the fit constrained each center to within 1° of its initial position and each Gaussian width to between 0.01° and 1° . Nonlinear least squares was performed with each count weighted by its Poisson uncertainty, $\sqrt{\max(I, 1)}$. The fitted center μ_i gives the peak position $2\theta_i$; its uncertainty was obtained from the corresponding σ_i standard deviation. Fit residuals and reduced χ^2 were also calculated to assess the fit.

For each fitted peak, the interplanar spacing was calculated from Bragg's law (equation 1) using first-order diffraction and $\lambda = 1.5406 \text{ \AA}$ for a K_α source. The uncertainty in d was propagated from the fitted uncertainty in 2θ . Given that multiple plots were available per sample, weighted means and uncertainties were calculated separately for each peak position and spacing. The values of $1/d^2$ and their propagated uncertainties were then calculated, and ratios (often relative to the first peak, as was the case with all cubics) were compared with the expected sequences for candidate crystal systems and Miller indices. Candidate indices and

lattice parameters were assessed for consistency with the measured spacings; Bravais-lattice selection rules (Appendix 6.1) were used to assess which reflections are allowed.

3 Results

3.1 Analysis of Copper, Nickel, and alloys

Figure 2 shows the summed gaussians fitted to the diffraction pattern for a Cu sample as an example. All other such fits are available in 6.2. A χ^2 value of 2.477 was obtained, showing that the results are statistically probable and reproducible. From the means of the gaussians and Bragg's law the inter-planar distances were found, and then averaged for all Cu sample runs. Angles for this particular run are shown in Table 1, along with the associated mean inter-planar distances for Cu. Uncertainties were calculated as explain in 2.2. We first try cubic systems, so a guess is made that the first peak occurs at (100), thus $1/d^2 = 1/a^2$. Dividing $1/d^2$ by $1/a^2$ for every d does not yield integers. However we find that it is an integer times a factor of three. Hence we multiply by three and find integers for $h^2 + k^2 + l^2$. From these integer we infer h, k, l , and hence the Miller index for each plane. Table 2 show the values obtained following this procedure.

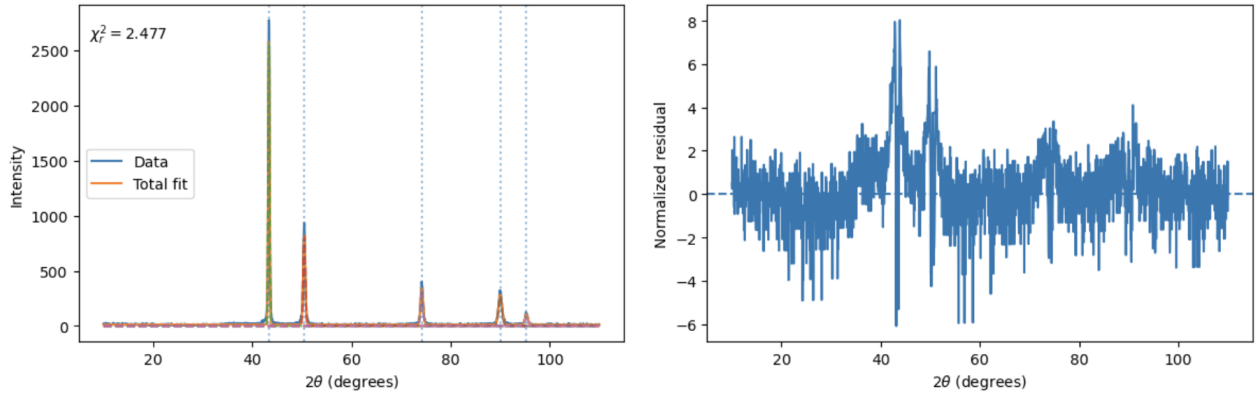


Figure 2: Plot of intensity against 2θ for the Copper sample. the summed gaussians (orange) are fitted to the data (blue).

2θ (deg)	d_{mean} (Å)
$43.3664 \pm 8 \cdot 10^{-4}$	$2.08485 \pm 4 \cdot 10^{-5}$
$50.489 \pm 2 \cdot 10^{-3}$	$1.80619 \pm 6 \cdot 10^{-5}$
$74.205 \pm 3 \cdot 10^{-3}$	$1.27694 \pm 5 \cdot 10^{-5}$
$90.017 \pm 4 \cdot 10^{-3}$	$1.08921 \pm 4 \cdot 10^{-5}$
$95.263 \pm 7 \cdot 10^{-3}$	$1.04260 \pm 6 \cdot 10^{-5}$

Table 1: Angle of the peaks and the associated inter-planar distances

$1/d^2$ (Å ⁻²)	$\frac{1/d^2}{a^2} \cdot 3$	$h^2 + k^2 + l^2$	index
$0.230064 \pm 8 \cdot 10^{-6}$	3.00 ± 0.00	3	(111)
$0.30653 \pm 2 \cdot 10^{-5}$	$3.9971 \pm 3 \cdot 10^{-4}$	4	(200)
$0.61329 \pm 4 \cdot 10^{-5}$	$7.9971 \pm 6 \cdot 10^{-4}$	8	(220)
$0.84291 \pm 6 \cdot 10^{-5}$	$10.9913 \pm 8 \cdot 10^{-4}$	11	(311)
$0.9200 \pm 1 \cdot 10^{-4}$	$11.996 \pm 1 \cdot 10^{-3}$	12	(222)

Table 2: Values obtained when attempting to index the Miller indices.

Based on the selection rules for cubic systems, we find that Copper is a face centered cubic (FCC). A similar method was applied to determine the same for Nickel, and all alloys of the two materials. The lattice parameters were then calculated and are shown in table 3 below, compared with theoretical values.

Sample	a_{exp} (Å)	a_{theory} (Å)	Δa (Å)
Ni	3.5223 ± 0.0001	3.5238	-0.0015 ± 0.0001
Cu	3.6126 ± 0.0001	3.6149	-0.0023 ± 0.0001
Cu ₂₅ Ni ₇₅	3.5415 ± 0.0002	3.5467	-0.0052 ± 0.0002
Cu ₅₀ Ni ₅₀	3.5629 ± 0.0003	3.5696	-0.0067 ± 0.0003
Cu ₇₅ Ni ₂₅	3.5864 ± 0.0002	3.5924	-0.0060 ± 0.0002

Table 3: Experimental and theoretical lattice parameters for Cu, Ni pure samples and alloys. The theoretical values for the alloys are calculated using Vegard's law .

For cubics, a viable alternate method for comparison with expected sequences is to plot the experimental values against the expected sequence of $(h^2 + k^2 + l^2)$ values for each of all 3 types of cubic Bravais lattices, which yielded the same results as indexing and comparing for all copper and nickel samples (alloys included) Example such plots are shown below in figure 3 for mean values of d^{-2} .

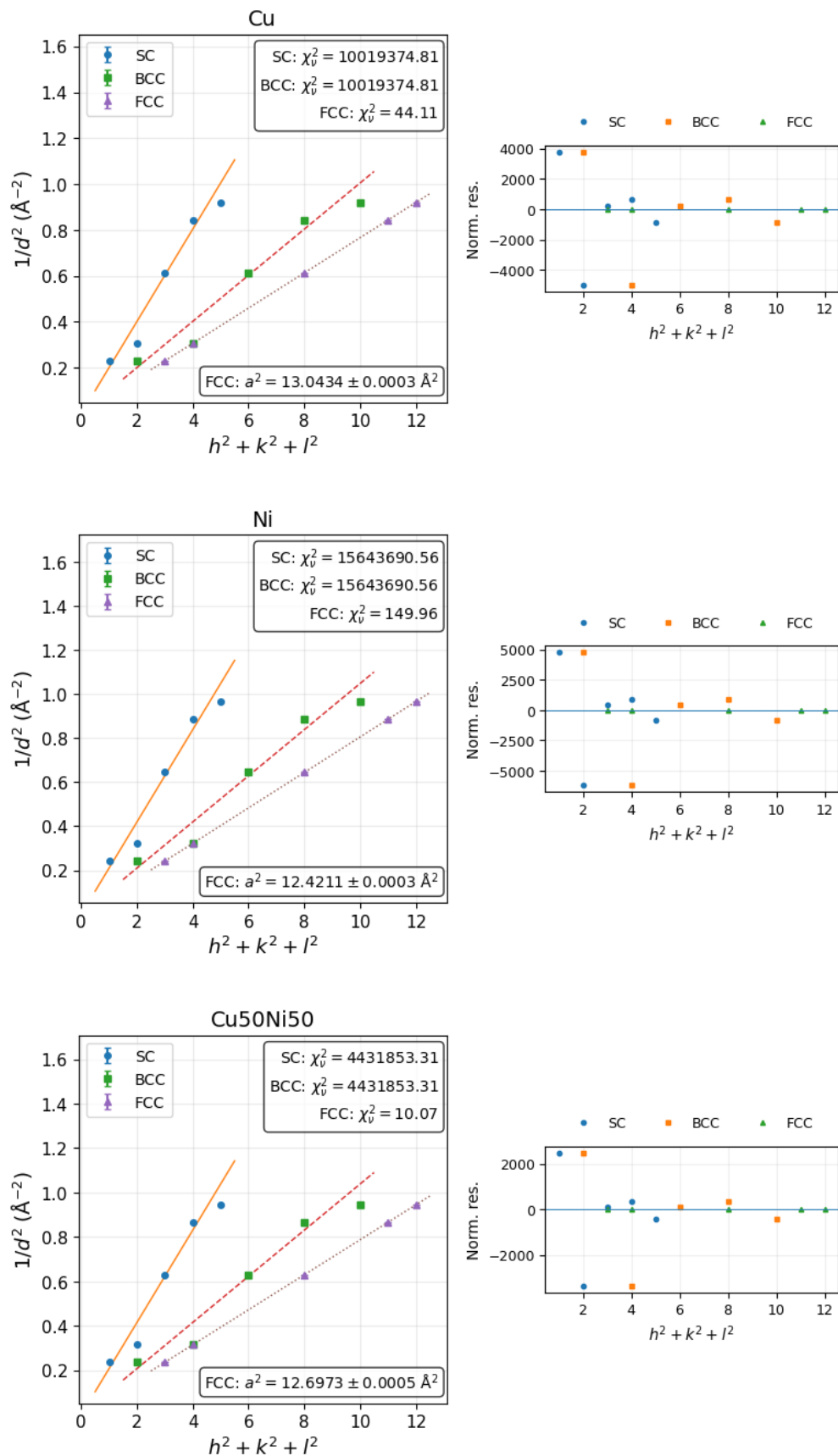


Figure 3: Determination of Cubic crystal structure and lattice parameter for Cu and Ni.

3.2 Lead, Tin, and alloys

The lead and tin samples' analysis proved to be more challenging, as the peaks were greater in number and not as well defined. Still, proper tweaking of the peak detection parameters lead to accurate identification of the peak angles, which were then used to calculate the interplanar distances as with Cu and Ni. Lead (Pb) followed a pattern similar to that of Cu and Ni, though showing many more peaks; it was indexed as another face-centered cubic (FCC) system, with a lattice parameter of $a_{Pb} = 4.94343 \pm 0.0001 \text{ \AA}$. As it is repetitive, the indexing of the lead sample is not shown here, but is available in the supplementary material. The tin sample required more in-depth analysis, as it did not match any cubic crystal structure. It was eventually identified to be a body-centered tetragonal, with parameters $a_{Sn} = 5.8278 \pm 0.0010 \text{ \AA}$ and $c_{Sn} = 3.1817 \pm 0.0008 \text{ \AA}$. As for the alloys, time proved insufficient to complete that part of the analysis, and no lattice parameter values were obtained, but the data is discussed below. Indexing of samples containing Sn is too bulky to be shown here, but is available in the supplementary material. Further discussion of the analysis of these samples is done in section 4.

3.3 Mystery Samples A and B

Analysis of the mystery samples provided was actually quite straightforward. Building upon the methodology developed for Copper and Nickel, we quickly found sample A to be a cubic system with a face-centered lattice, with a ratio sequence and Miller indices identical to that of the Copper and Nickel samples; sample B proved to be a cubic system with a primitive lattice. Again, as it is repetitive, the indexing of sample A is not shown here, but is available in the supplementary material. The indexing of sample B is shown in table 4 below, where a is the first inreplanar distance, as is the case with PC. The lattice parameters were calculated to be $a_A = 3.6110 \pm 0.0001 \text{ \AA}$ and $a_B = 3.7482 \pm 0.0004 \text{ \AA}$ respectively.

$1/d^2$ ($\text{\AA}^{-2}$)	$\frac{1/d^2}{a^2}$	Expected PC ratio	Miller indices
0.071179 ± 0.000016	1.00 ± 0.00	1	(100)
0.142535 ± 0.000020	2.00248 ± 0.00054	2	(110)
0.213807 ± 0.000010	3.00379 ± 0.00071	3	(111)
0.285031 ± 0.000021	4.00442 ± 0.00097	4	(200)
0.356334 ± 0.000050	5.0062 ± 0.0014	5	(210)
0.427618 ± 0.000065	6.0076 ± 0.0017	6	(211)
0.570263 ± 0.000036	8.0117 ± 0.0019	8	(220)
0.64134 ± 0.00011	9.0102 ± 0.0026	9	(221), (300)
0.71252 ± 0.00016	10.0103 ± 0.0033	10	(310)
0.784099 ± 0.000050	11.0159 ± 0.0026	11	(311)
0.85524 ± 0.00011	12.0153 ± 0.0032	12	(222)
0.92726 ± 0.00026	13.0271 ± 0.0048	13	(320)
0.99755 ± 0.00018	14.0146 ± 0.0041	14	(321)

Table 4: XRD peak indexing for sample B.

4 Discussion

The CuNi samples provide the clearest test of the peak-indexing procedure. The measured Cu reflections are consistent with the sequence (111), (200), (220), (311), and (222), and satisfy the FCC selection rule that the Miller indices are either all odd or all even. The corresponding cubic fits give the same FCC sequences for Ni and for the Cu–Ni alloys. The measured lattice parameter increases with Cu content, from 3.5223 ± 0.0001 Å for Ni to 3.6126 ± 0.0001 Å for Cu, which is in close agreement with the linear Vegard-law estimates listed in Table 3.

All five experimental lattice parameters in that table are lower than their theoretical or Vegard-law values. The differences range from 0.0015 to 0.0067 Å (less than 0.2%), but are larger than the quoted fit-based uncertainties. This means those small uncertainties describe the precision estimated from the fitted peaks and repeat measurements rather the

full accuracy of the lattice parameters, which is a flaw in our method. The reported Cu fit statistic of $\chi^2 = 2.477$ should likewise be interpreted together with its degrees of freedom and the precise definition of the statistic; it does not on its own demonstrate reproducibility, but the other Cu runs shown in the appendix, and the similarity between runs for each sample, also show in 6.2 do. In the end, the CuNi results do show that the combination of Bragg’s law, Miller-index sequences, and lattice selection rules can identify structures and estimate lattice parameters from measured diffraction peaks, which was the primary goal of this preliminary study.

The Pb and Sn results show the importance of comparing candidate crystal systems rather than assuming that every sample is cubic. The Pb pattern was found as FCC, with $a_{\text{Pb}} = 4.94343 \pm 0.0001 \text{ \AA}$, while the Sn one did not match a cubic sequence and was shown to be a body-centered tetragonal structure, with $a_{\text{Sn}} = 5.8278 \pm 0.0010 \text{ \AA}$ and $c_{\text{Sn}} = 3.1817 \pm 0.0008 \text{ \AA}$. This is also in close accordance with theoretical values. The PbSn alloys were not fully analyzed, but the measured peaks are consistent with a tetragonal structure, which is expected for a mixture of FCC and BCT materials.

The two mystery patterns were assigned to cubic structures: sample A to FCC and sample B to primitive cubic (PC), based on their observed ratio sequences and the applicable selection rules. These assignments describe the diffraction patterns; the results presented here do not independently establish the chemical identities of the samples. However, it does strongly suggest that is the case.

5 Conclusion

6 Appendix

6.1 Inter-planar distance, Bravais Lattices and Miller indices

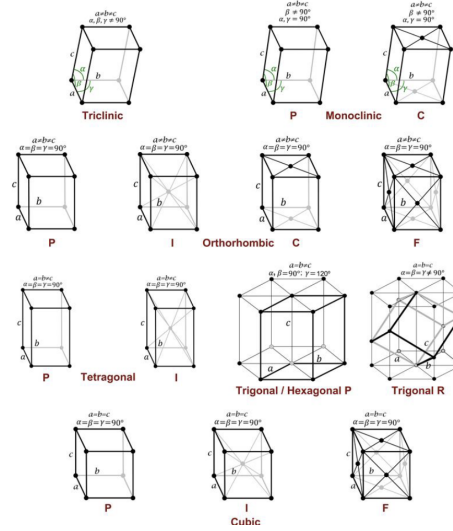


Figure 4: Fourteen Bravais lattices and their seven associated groups.

The selection rules for cubic systems are as follows:

- PC: all h, k, l allowed
- FCC: all even or all odd h, k, l allowed
- BCC: $h + k + l$ even

For tetragonal systems, the selection rules are as follows:

- P: all h, k, l allowed
- I: $h + k + l$ even

For the following equations: d is the inter planar distance, a, b, c the inter-atomic distances, h, k, l the indices for a given Miller index (hkl).

Cubic:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Tetragonal:

$$\frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

Hexagonal:

$$\frac{1}{d^2} = \frac{4}{3} \left(\frac{h^2 + k^2 + hk}{a^2} \right) + \frac{l^2}{c^2}$$

Orthorhombic:

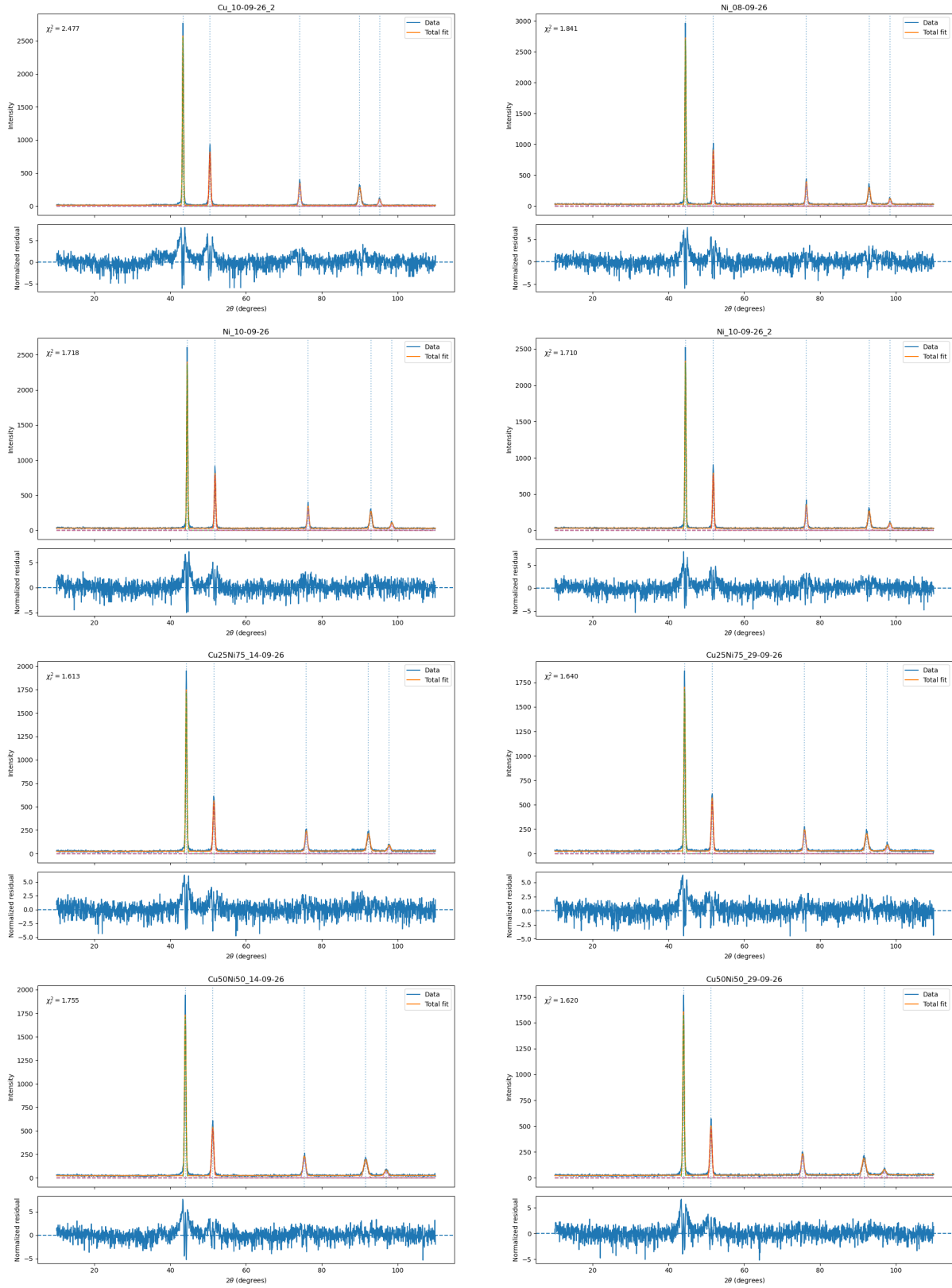
$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Vegard's law:

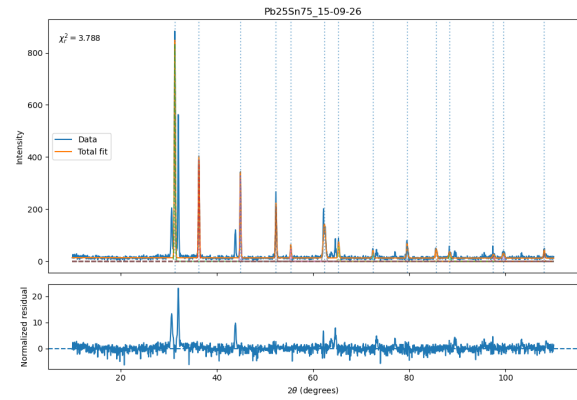
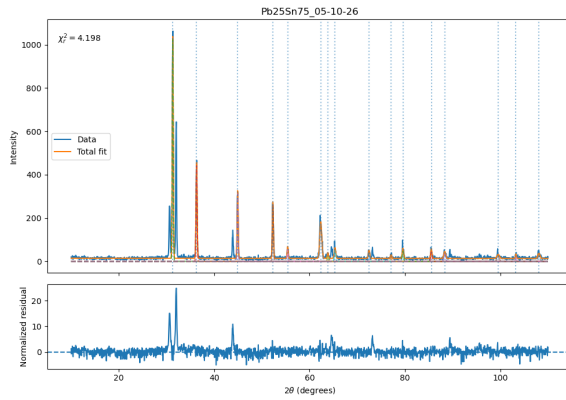
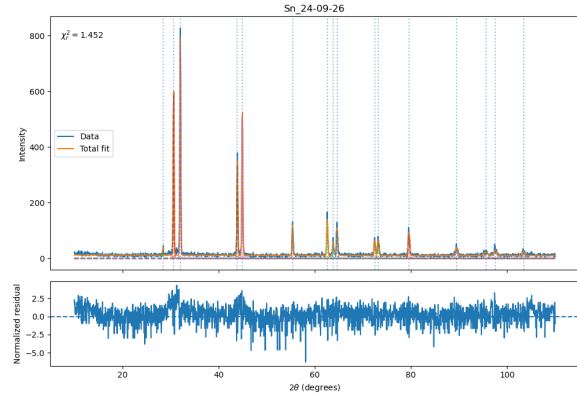
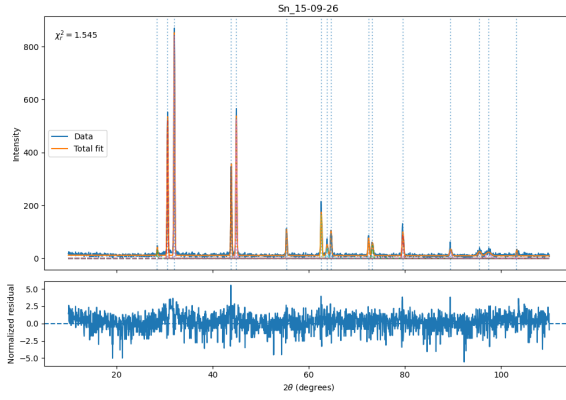
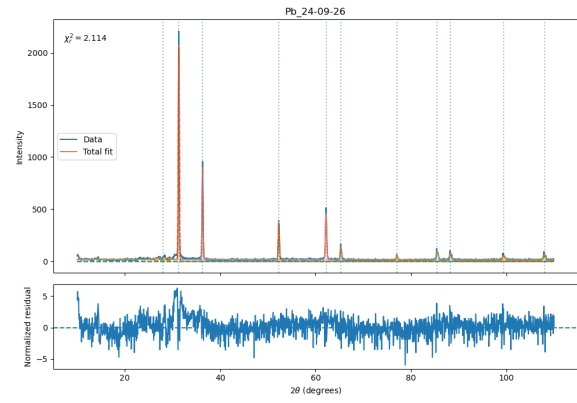
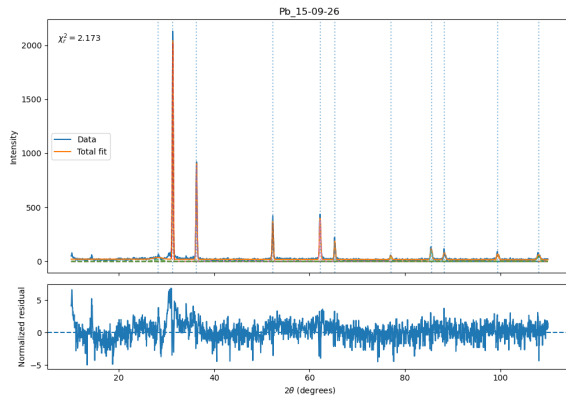
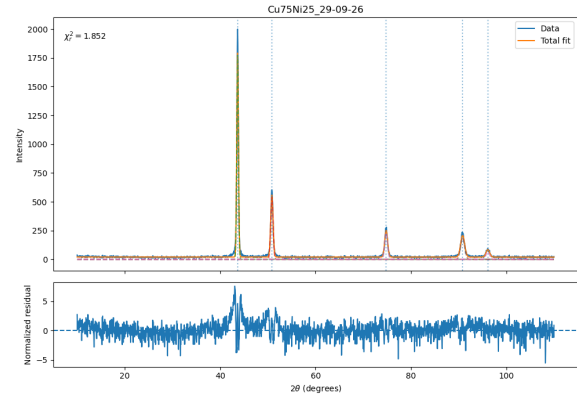
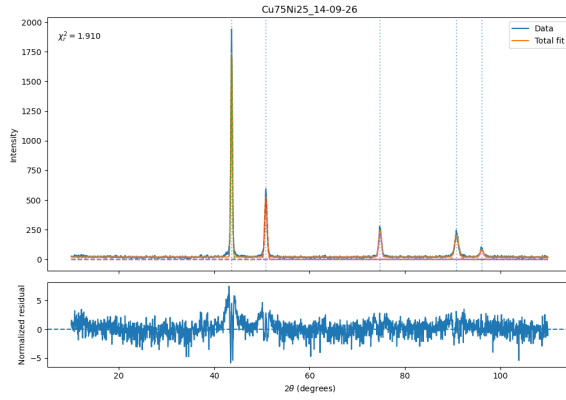
$$a_{\text{alloy}} = xa_{Cu} + (1 - x)a_{Ni}$$

6.2 Supplementary Figures and Tables

Gaussian fits and normalized residuals (1/3)



Gaussian fits and normalized residuals (2/3)



Gaussian fits and normalized residuals (3/3)

